

Formelsamling för kursen Kvantmekanik och Spektroskopi

Detta dokument innehåller alla formler som ni har tillgängliga under tentan. Värden på alla konstanter finns i *Book of Data*. Övriga formler som behövs finns angivna i uppgiftstexten.

1 Allmänna formler

Energi, frekvens, våglängd, vågtal

$$E = h\nu = \hbar\omega = hc\tilde{\nu}, \quad \Delta E = h\nu$$

$$\lambda\nu = c, \quad \tilde{\nu} = \frac{1}{\lambda}, \quad \omega = 2\pi\nu$$

De Broglie och kinetik

$$\lambda = \frac{h}{p}, \quad p = mv, \quad E_k = \frac{p^2}{2m} = \frac{1}{2}mv^2$$

Fotoelektriska effekten

$$h\nu = \Phi + E_k$$

Reducerad massa

$$\mu = \frac{m_A m_B}{m_A + m_B}$$

Allmänt superpositionstillstånd

$$\psi = \sum_j c_j \psi_j, \quad \hat{A}\psi = \psi'$$

Kommutator

$$[\hat{A}, \hat{B}] \equiv \hat{A}\hat{B} - \hat{B}\hat{A}, \quad [\hat{A}, \hat{B}] = -[\hat{B}, \hat{A}]$$

$$[\hat{A}, \hat{B} + \hat{C}] = [\hat{A}, \hat{B}] + [\hat{A}, \hat{C}], \quad [c\hat{A}, \hat{B}] = c[\hat{A}, \hat{B}] = [\hat{A}, c\hat{B}]$$

Hermiteisk operator

$$\hat{A}^\dagger = \hat{A}, \quad (\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger, \quad (\hat{A} + \hat{B})^\dagger = \hat{A}^\dagger + \hat{B}^\dagger$$

$$\int (\psi_j)^* \hat{A} \psi_k \, d\tau = \left(\int (\psi_k)^* \hat{A} \psi_j \, d\tau \right)^*$$

Normalisering och ortonormalitet

$$\int |\psi|^2 \, d\tau = 1, \quad \int (\psi_j)^* \psi_k \, d\tau = \delta_{jk}, \quad v_j^\dagger v_k = \delta_{jk}$$

Egenekvation

$$\hat{A}\psi_k = a_k \psi_k \quad (\text{alt. } \hat{A}|k\rangle = a_k|k\rangle)$$

Väntevärde

$$\langle \hat{A} \rangle = \sum_j |c_j|^2 a_j = \int \psi^*(\mathbf{r}) \hat{A} \psi(\mathbf{r}) \, d\tau = \psi^\dagger \hat{A} \psi$$

Positions- och rörelsemängdsoperator (1D, riktning q)

$$\hat{q} = q, \quad \hat{p}_q = -i\hbar \frac{d}{dq}, \quad [\hat{q}, \hat{p}_q] = i\hbar$$

Osäkerhetsrelationer

$$\Delta q \Delta p_q \geq \frac{\hbar}{2}, \quad \Delta A \Delta B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|$$

Lambert–Beers lag

$$\log_{10} \left(\frac{I_0}{I_t} \right) = \epsilon c l = A$$

Komplexa tal

$$i^2 = -1, \quad \frac{1}{i} = -i, \quad z^* = \text{konjugat}, \quad z z^* = |z|^2$$

$$z + z^* = 2\Re(z), \quad z - z^* = 2i\Im(z)$$

Hamiltonoperator & Schrödingerekvationen

Generellt

$$\hat{\mathcal{H}} = \hat{\mathcal{H}}_k + \hat{\mathcal{H}}_p, \quad \hat{\mathcal{H}}_p = V(q)$$

Kinetisk energi

$$\hat{\mathcal{H}}_{k,1D} = \frac{\hat{p}_q^2}{2m} = -\frac{\hbar^2}{2m} \frac{d^2}{dq^2}, \quad \hat{\mathcal{H}}_{k,3D} = -\frac{\hbar^2}{2m} \nabla^2$$

Laplaceoperatorn i sfäriska koordinater

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}$$

Tidsberoende Schrödingerekvationen

$$i\hbar \frac{\partial \psi(\mathbf{r}, t)}{\partial t} = \hat{\mathcal{H}} \psi(\mathbf{r}, t)$$

$$\text{Separation av variabler: } \psi(\mathbf{r}, t) = \psi(\mathbf{r}) e^{-iEt/\hbar}$$

Tidsberoende Schrödingerekvationen ($\hat{\mathcal{H}}\psi = E\psi$)

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi(x)}{dx^2} + V(x) \psi(x) = E \psi(x), \quad -\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{r}) + V(\mathbf{r}) \psi(\mathbf{r}) = E \psi(\mathbf{r})$$

Harmonisk oscillator

$$V(x) = \frac{k}{2} x^2, \quad x = R - R_e, \quad (\text{ofta } m \rightarrow \mu \text{ för diatom})$$

Partikel på ring / sfär

$$\hat{\mathcal{H}}_{\text{ring}} = -\frac{\hbar^2}{2I} \frac{d^2}{d\phi^2}, \quad \hat{\mathcal{H}}_{\text{sfär}} = -\frac{\hbar^2}{2I} \nabla_\Omega^2$$

H-liknande atom (Coulomb)

$$V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}, \quad V_{\text{eff}}(r) = -\frac{Ze^2}{4\pi\epsilon_0 r} + \frac{\hbar^2 l(l+1)}{2\mu r^2}$$

$$\hat{\mathcal{H}}_k \text{ (exakt)} = -\frac{\hbar^2}{2m_n} \nabla_n^2 - \frac{\hbar^2}{2m_e} \nabla_e^2, \quad \hat{\mathcal{H}}_k \text{ (approx)} \approx -\frac{\hbar^2}{2\mu} \nabla_{\text{rel}}^2$$

3 Translationsrörelse

Fri partikel ($V = 0$)

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} = E \psi, \quad \psi(x) = A e^{ikx} + B e^{-ikx}, \quad E = \frac{\hbar^2 k^2}{2m}$$

Partikel i 1D-låda

$$n\lambda = 2L, \quad \psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \quad E_n = \frac{n^2 \hbar^2}{8mL^2}$$

Partikel i 3D-låda

$$\psi_{n_1 n_2 n_3} = \sqrt{\frac{8}{L_1 L_2 L_3}} \sin\left(\frac{n_1 \pi x}{L_1}\right) \sin\left(\frac{n_2 \pi y}{L_2}\right) \sin\left(\frac{n_3 \pi z}{L_3}\right)$$

$$E_{n_1 n_2 n_3} = \frac{\hbar^2}{8m} \left(\frac{n_1^2}{L_1^2} + \frac{n_2^2}{L_2^2} + \frac{n_3^2}{L_3^2} \right)$$

4 Rotation & angular momentum

Klassiskt (partikel på ring)

$$J = \pm pr, \quad J_z = p_z r, \quad E_k = \frac{p_z^2}{2m} = \frac{J_z^2}{2I}, \quad I = mr^2$$

Partikel på ring

$$\psi_{m_l}(\phi) = \frac{1}{\sqrt{2\pi}} e^{im_l \phi}, \quad E_{m_l} = \frac{\hbar^2 m_l^2}{2I}, \quad m_l = 0, \pm 1, \pm 2, \dots$$

Partikel på sfär (sfärharmoniska)

$$Y_{l,m_l}(\theta, \phi) = \Theta_l(\theta) \Phi_{m_l}(\phi), \quad E_l = \frac{\hbar^2}{2I} l(l+1)$$

$$l = 0, 1, \dots, \quad m_l = 0, \pm 1, \dots, \pm l$$

$$Y_{l,m_l}^*(\theta, \phi) = (-1)^{m_l} Y_{l,-m_l}(\theta, \phi), \quad \int Y_{l_1}^{m_1*} Y_{l_2}^{m_2} d\Omega = \delta_{l_1 l_2} \delta_{m_1 m_2}$$

Atomorbital

$$\psi_{n,l,m_l}(r, \theta, \phi) = R_{n,l}(r) Y_{l,m_l}(\theta, \phi)$$

Orbitaloperatorer

$$\hat{L}_x = -i\hbar \left(y \frac{d}{dz} - z \frac{d}{dy} \right), \quad \hat{L}_y = -i\hbar \left(z \frac{d}{dx} - x \frac{d}{dz} \right)$$

$$\hat{L}_z = -i\hbar \left(x \frac{d}{dy} - y \frac{d}{dx} \right) = -i\hbar \frac{d}{d\phi}$$

$$\hat{L}_z Y_{l,m_l} = m_l \hbar Y_{l,m_l}, \quad \hat{L}^2 Y_{l,m_l} = l(l+1) \hbar^2 Y_{l,m_l}, \quad \hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$$

Allmänt för angular momentum ($\hat{J} = \hat{L}$ eller $\hat{J} = \hat{I}$)

$$[\hat{J}^2, \hat{J}_q] = 0, \quad [\hat{J}_x, \hat{J}_y] = i\hbar \hat{J}_z \text{ (cykliskt)}, \quad |\mathbf{J}| = \hbar \sqrt{J(J+1)}$$

Spinn 1/2 och Pauli-matriser ($I = s = 1/2$)

$$\psi\{\alpha\} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \psi\{\beta\} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \psi = c_\alpha \psi\{\alpha\} + c_\beta \psi\{\beta\}$$

$$I_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad I_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad I_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

5 Rotationspektroskopi

Tröghetsmoment

$$I = \sum_j m_j x_j^2, \quad \text{specialfall: } I = MR^2, \quad M = m \text{ eller } M = \mu$$

Klassisk energi (asymmetrisk rotor)

$$E_k = \frac{J_a^2}{2I_a} + \frac{J_b^2}{2I_b} + \frac{J_c^2}{2I_c}$$

Linjär rotor

$$E_J = \frac{\hbar^2}{2I} J(J+1) = hB J(J+1), \quad J = 0, 1, 2, \dots$$

$$B = \frac{\hbar}{4\pi I} \text{ (Hz)}, \quad \tilde{B} = \frac{\hbar}{4\pi c I} \text{ (m}^{-1}\text{)}, \quad \tilde{D} = \frac{4\tilde{B}^3}{\tilde{\nu}^2}$$

$$\tilde{F}(J) = \tilde{B}J(J+1), \quad \tilde{\nu}_J = \tilde{F}(J+1) - \tilde{F}(J) = 2\tilde{B}(J+1)$$

$$\tilde{F}(J) = \tilde{B}J(J+1) - \tilde{D}J^2(J+1)^2, \quad \tilde{\nu}_J = 2\tilde{B}(J+1) - 4\tilde{D}(J+1)^3$$

$$N_J \propto (2J+1) \exp\left(-\frac{E_J}{k_B T}\right)$$

Symmetrisk rotor

$$\tilde{F}(J, K) = \tilde{B}J(J+1) + (\tilde{A} - \tilde{B})K^2, \quad K = 0, \pm 1, \dots, \pm J$$

$$\tilde{B} = \frac{\hbar}{4\pi c I_\perp}, \quad \tilde{A} = \frac{\hbar}{4\pi c I_\parallel}$$

6 Vibration: harmonisk oscillator

$$E_v = \hbar\omega \left(v + \frac{1}{2} \right) = hc \tilde{\nu} \left(v + \frac{1}{2} \right), \quad v = 0, 1, 2, \dots$$

$$\omega = \sqrt{\frac{k}{\mu}}, \quad \tilde{\nu} = \frac{1}{2\pi c} \sqrt{\frac{k}{\mu}}, \quad \tilde{G}(v) = \tilde{\nu} \left(v + \frac{1}{2} \right) \Rightarrow \Delta \tilde{G}_{v+1/2} = \tilde{\nu}$$

$$\tilde{S}(v, J) = \tilde{F}(J) + \tilde{G}(v) = \tilde{\nu} \left(v + \frac{1}{2} \right) + \tilde{B}J(J+1)$$

För $\Delta v = 1$ erhålls tre grenar:

$$\Delta J = 0: \quad \tilde{\nu}_Q(J) = \tilde{\nu}, \quad \Delta J = \pm 1: \quad \tilde{\nu}_{R/P}(J) = \tilde{\nu} \pm 2\tilde{B}J$$

Antal vibrationsmoder: linjär $3N - 5$, icke-linjär $3N - 6$.

7 Vibration: anharmonisk oscillator

Morsepotential

$$V(x) = hc \tilde{D}_e \left(1 - e^{-ax} \right)^2, \quad a = \left(\frac{\mu\omega^2}{2hc \tilde{D}_e} \right)^{1/2}, \quad x_e = \frac{\tilde{\nu}}{4\tilde{D}_e}$$

$$\tilde{G}(v) = \tilde{\nu} \left(v + \frac{1}{2} \right) - x_e \tilde{\nu} \left(v + \frac{1}{2} \right)^2, \quad \Delta \tilde{G}_{v+1/2} = \tilde{\nu} - 2(v+1)x_e \tilde{\nu}$$

8 Atomstruktur & elektronspektroskopi

H-liknande atom

$$E_n = -\frac{\mu Z^2 e^4}{32\pi^2 \epsilon_0^2 \hbar^2} \frac{1}{n^2}, \quad Z = 1: \quad E_n = -\frac{hc \tilde{R}_H}{n^2}, \quad \tilde{\nu} = \tilde{R} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\tilde{R}_H = \frac{\mu_H e^4}{8\epsilon_0^2 \hbar^3 c}, \quad \tilde{R}_\infty = \frac{m_e e^4}{8\epsilon_0^2 \hbar^3 c}$$

Bohr-radie

$$a_0 = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2}$$

Radiell sannolikhetsfördelning

$$P(r) dr = r^2 |R(r)|^2 dr$$

Orbitalapproximationen och spinn-orbitaler

$$\Psi(r_1, \dots, r_N) = \psi(r_1) \psi(r_2) \dots \psi(r_N) + \dots$$

$$\psi_\phi^\alpha(j) = \phi(j) \alpha(j), \quad \psi_\phi^\beta(j) = \phi(j) \beta(j)$$

Effektiv kärnladdning

$$Z_{\text{eff}} = Z - \sigma$$

Koppling (Clebsch–Gordan)

$$L = |l_1 - l_2|, |l_1 - l_2| + 1, \dots, l_1 + l_2, \quad M_L = 0, \pm 1, \dots, \pm L$$

$$J_{\text{tot}} = |j_1 - j_2|, |j_1 - j_2| + 1, \dots, j_1 + j_2, \quad M_J = 0, \pm 1, \dots, \pm J_{\text{tot}}$$

J betecknar totala angular momentum från koppling $\{L, S\} \rightarrow J$

Spin-orbit (form)

$$E_{\text{SOC}} \propto Z^4 [J(J+1) - L(L+1) - S(S+1)]$$

9 NMR

$$\nu_0 = \frac{\omega_0}{2\pi} = -\frac{\gamma}{2\pi} B_0, \quad \nu_{\text{res}} = -\frac{\gamma}{2\pi} B_{\text{sum}} = \nu_0(1 + \delta_j)$$

$$\delta_j = \frac{\nu_j - \nu_{\text{ref}}}{\nu_{\text{ref}}}, \quad \delta_{\text{ppm}} = \delta_j \times 10^6 \text{ ppm}$$

$$a(t) = M_0 \left(1 - 2e^{-t/T_1} \right), \quad a(t) \propto M_0 e^{-t/T_2}$$

$$\Delta\nu_{1/2} = 2\lambda = \frac{1}{\pi T_2}$$